Global-from-Local Restart in Lean 4: Conditional Continuation Theorem Prohibiting Finite-Time Termination Under Uniform Ceiling Bounds

Benjamin Stanley Frohman
ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/global-from-local-restart
Dedicated 1-to-1 Zenodo DOI: 10.5281/zenodo.23653003

Earliest Commit Provenance: 2026-10-07 04:10:30 AM CDT (UTC-5)
Siloed Archival Edition: October 25, 2026 (America/Chicago CDT)

Abstract

Standalone zero-sorry Lean 4 formalization (‘Continuation.lean‘, ‘GlobalFromLocalRestart.lean‘) proving ‘Continuation.no_finite_last_time_of_restart‘: an abstract continuation theorem showing that any evolution system satisfying local existence with uniform step size below a ceiling, a priori ceiling preservation, and uniqueness cannot terminate at a finite maximal time T*.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Proves the abstract conditional continuation lemma in Lean 4 with zero sorry; explicitly states that applying it to 3D Navier-Stokes requires supplying the a priori ceiling hypothesis independently.

- **Repository:** `https://github.com/BenFrohman/global-from-local-restart`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23653003`
- **Earliest Git Commit SHA:** `0aa100b6be5851bd0b73ed51e2ae03078bfa6a46` (2026-10-07 04:10:30 AM CDT (UTC-5))
- **License & Copyright:** Copyright © 2026 Benjamin Stanley Frohman. Licensed under the Apache License, Version 2.0.

2 Technical Overview & Repository Specification

Copyright (c) 2026 Benjamin Stanley Frohman. All rights reserved.
Author: Benjamin Stanley Frohman
License: Apache-2.0. See ‘LICENSE‘, ‘NOTICE‘, and ‘COPYRIGHT.md‘.
‘Continuation.no_finite_last_time_of_restart‘ is an induction. If time 0 is reached and every reached time extends by a positive step, then there is no finite last reached time.
‘Continuation.global_of_local_restart‘ is that induction with the step taken from a ‘LocalWitness‘. The witness is an argument. It is not constructed from smooth data.
This file does not contain a uniqueness hypothesis. It does not derive a cubic

sink. It is not an unconditional Navier-Stokes theorem.
Pinned toolchain: ‘leanprover/lean4:v4.30.0-rc1‘.
“‘bash
git clone https://github.com/BenFrohman/global-from-local-restart.git
cd global-from-local-restart
lake build
“‘

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/global-from-local-restart` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.json`) and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`AUTHORS.md`	103 B	`f1396b31be4df588...`
`COPYRIGHT.md`	313 B	`8715875de7652901...`
`Continuation.lean`	1881 B	`e66c894aee3cbf99...`
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	676 B	`86df144394d21f55...`
`README.md`	883 B	`96206f7c9343648a...`
`lakefile.lean`	363 B	`87efb1784f4ab7c9...`
`lean-toolchain`	29 B	`1098ca8368e0db12...`

Attribution & Licensing Notice: Copyright © 2026 Benjamin Stanley Frohman (ORCID: 0009-0006-7068-3718). Licensed under the Apache License, Version 2.0.

Lean declaration	`#print axioms`

A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  https://github.com/BenFrohman/global-from-local-restart
- Verified commit:
  `5831a82722a5b4a0d18f960b03afb5105dc0f57f`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  https://github.com/BenFrohman/global-from-local-restart/actions/runs/41568066665
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  https://github.com/BenFrohman/global-from-local-restart/archive/5831a82722a5b4a0d18f960b03afb5105dc0f57f.tar.gz
  `87d7b447b08bb233cdd86dfe8347e33ebfe24ec6d10417ffbdee1d81a13fd0a7`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 3 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`Continuation.LocalWitness.T_pos`	`(no axioms)`
`Continuation.global_of_local_restart`	`(no axioms)`
`Continuation.no_finite_last_time_of_restart`	`(no axioms)`